maximize the probability of the empirical data. Before training, the generative model will assign low probability to examples taken from the training data set, and the objective is for the generative model to assign high probability to examples drawn from the data set. After a sufficient number of iterations, the generative model will have assigned high probability to the empirical data, and we can then sample from this distribution to generate new, synthetic data.

Figure 7.17. This shows the predicted action-value distributions produced by an untrained Dist-DQN and the target distribution after observing a reward. There are three separate action-value distributions of length 51 elements, but here they've been concatenated into one long vector to illustrate the overall fit between the prediction and target. The first 51 elements correspond to the action-value distribution of the NO-OP operation, the second 51 elements correspond to the action-value distribution of the UP action, and the last 51 elements correspond to the DOWN distribution. You can see the prediction is a completely flat (uniform) distribution for all three actions, whereas the target distribution has a mode (a peak) for action 0 and some noisy peaks for the other two actions. The goal is to get the prediction to match the target distribution.

Figure 7.18. Top: The concatenated action-value distributions for all three actions after training. Bottom: Loss plot over training time. The baseline loss is decreasing, but we see ever-increasing spikes.

Figure 7.19. Each row contains the action-value distributions for an individual state, and each column in a row is the distribution for actions 0, 1, and 2 respectively.

Figure 7.20. The loss plot for training Dist-DQN on the Atari game Freeway. The loss gradually declines but has significant "spikiness" due to the periodic target network updates.

Figure 7.21. Left: Screenshot of live gameplay in Atari Freeway. Right: The corresponding action-value distributions of each of the each actions overlaid. The spike on the right corresponds to the UP action and the spike on the left corresponds mostly to the NO-OP action. Since the right spike is larger, the agent is more likely to take the UP action, which seems like the right thing to do in this case. It is difficult